

The Doppler Operator Generated from Its Seed: Explicit Theorems and Closed Forms

Every $\hat{\mathcal{D}}_{\ell\ell'\ell''}^m$ from the $\hat{\mathcal{D}}_{000}$ base case — no kernels, no integrals, no boosts

rSZ / Doppler-operator programme · 9 July 2026 · companion script `doppler_generator.w1` (log
out_doppler_generator.txt)

Third document of the series; the two companions prove the kernel-level inputs (*Rapidity ODE*: Lemmas 1–4, N3; *Index Recurrences*: Master Lemma, Theorem U, corrections).

Summary. The companion documents proved the raising machinery at the level of the aberration kernel. Here everything is restated *at the level of the Doppler operator itself*, as a closed, self-contained calculus: a seed family (G0), four solved ladders (G1), three symmetries (G2), and a generation theorem (G3) by which **any element $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q)$ is computed in closed form from the $\hat{\mathcal{D}}_{000}$ base case alone**, by finitely many additions and multiplications by γ , $1/p$ and rational constants. No kernels, boost operators, integrals, or derivatives appear in the rules. The scheme is implemented and verified exactly — the generated closed forms equal the defining operator as identities at symbolic q and symbolic rapidity for all indices ≤ 3 — and a catalog of closed forms is produced, including elements not published on the site. A by-product is a new *mirror rapidity ODE* (G5b) in which only the middle indices move. Section 6 goes further and removes the auxiliary family from the raising relations themselves: Theorems P0–P4 are index raising/lowering relations in which **every term is a physical Doppler element $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q+n)$** — the outer pair relation (P1), explicit axis ladders seeded by $\hat{\mathcal{D}}_{000}$ (P0), a rank-one theorem (P2), and the pure-physical middle-index step (P3) — together a generation scheme whose every intermediate is physical (P4).

1 Objects

DEFINITION. All statements are for the extended Doppler family on the weight lattice $(j, k) \in \mathbb{Z}^2$ (spin $s = 0$; on ν^q ; m fixed throughout, $m = 0$ unless stated):

$${}^{(j,k)}\mathcal{X}_{\ell;\ell'_1,\ell'_2;\ell''}^m(q) = \frac{1}{\gamma} {}^{-1-j}\hat{\mathcal{B}}_{\ell\ell'_1}^m(\nu, -\beta) {}^{-k}\hat{\mathcal{B}}_{\ell'_2\ell''}^m(\nu, \beta) \Big|_{\nu^q},$$

with the **physical Doppler operator as the diagonal slice** $\hat{\mathcal{D}}_{\ell\ell'\ell''}^m(q) = {}^{(0,0)}\mathcal{X}_{\ell;\ell',\ell';\ell''}^m(q)$. The extension is forced: the middle index is shared between the two scattering steps, and no recursion closes on the diagonal slice alone (companion II, Remark 2). Constants:

$C_\ell^m = \sqrt{(\ell^2 - m^2)/(4\ell^2 - 1)}$ ($C_\ell^m = 0$ for $\ell \leq m$; at $m = 0$, $C_\ell = \ell/\sqrt{4\ell^2 - 1}$); rapidity $\beta = \tanh \eta$, $\gamma \pm p = e^{\pm\eta}$; and the single special function of the whole theory,

$$J_w = \frac{(\gamma + p)^w - (\gamma - p)^w}{w} = \frac{2 \sinh(w\eta)}{w}, \quad J_{-w} = J_w.$$

2 G0 — the seed family (the $\hat{\mathcal{D}}_{000}$ base case)

THEOREM G0 (SEEDS, $M = 0$).

$${}^{(j,k)}\mathcal{X}_{0;0,0;0}(q) = \frac{J_{2+j+q} J_{1+k+q}}{4\gamma p^2}, \quad \text{in particular} \quad \hat{\mathcal{D}}_{000}(q) = \frac{J_{2+q} J_{1+q}}{4\gamma p^2}. \quad (\text{G0})$$

The whole seed family is the $\hat{\mathcal{D}}_{000}$ closed form with integer-shifted J -arguments — by the gauge identity (G2c) it is literally $\hat{\mathcal{D}}_{000}$ at shifted q , applied factor-wise.

3 G1 — the four solved ladders

THEOREM G1 (RAISING RULES). Write $\mathcal{X}$ for ${}^{(j,k)}\mathcal{X}_{\ell;\ell'_1,\ell'_2;\ell''}^m(q)$ and show only what changes. Elements with any index below m are zero. Then:

$$\begin{aligned} \text{(G1-}\ell\text{)} \quad \mathcal{X}[\ell] &= +\frac{1}{C_\ell^m p} \left(\gamma \mathcal{X}[\ell-1] - {}^{(j+1,k)}\mathcal{X}[\ell-1] \right) - \frac{C_{\ell-1}^m}{C_\ell^m} \mathcal{X}[\ell-2], \\ \text{(G1-}\ell'_1\text{)} \quad \mathcal{X}[\ell'_1] &= -\frac{1}{C_{\ell'_1}^m p} \left(\gamma \mathcal{X}[\ell'_1-1] - {}^{(j-1,k)}\mathcal{X}[\ell'_1-1] \right) - \frac{C_{\ell'_1-1}^m}{C_{\ell'_1}^m} \mathcal{X}[\ell'_1-2], \\ \text{(G1-}\ell'_2\text{)} \quad \mathcal{X}[\ell'_2] &= -\frac{1}{C_{\ell'_2}^m p} \left(\gamma \mathcal{X}[\ell'_2-1] - {}^{(j,k+1)}\mathcal{X}[\ell'_2-1] \right) - \frac{C_{\ell'_2-1}^m}{C_{\ell'_2}^m} \mathcal{X}[\ell'_2-2], \\ \text{(G1-}\ell''\text{)} \quad \mathcal{X}[\ell''] &= +\frac{1}{C_{\ell''}^m p} \left(\gamma \mathcal{X}[\ell''-1] - {}^{(j,k-1)}\mathcal{X}[\ell''-1] \right) - \frac{C_{\ell''-1}^m}{C_{\ell''}^m} \mathcal{X}[\ell''-2]. \end{aligned}$$

The four rules commute; each references only elements with a strictly smaller raised index, at the same or a neighboring lattice point (the compass of Theorem U: east, west, north, south).

Proof. Solve each ladder of Theorem U (companion II) for its top element. (G1- ℓ) and (G1- ℓ'') are N5 and N6; (G1- ℓ'_1) and (G1- ℓ'_2) are the solved west and north ladders, whose diagonal-restricted product is the site's middle-index relation (M) — here they are kept separate, which is both simpler and strictly more general. All four are verified exactly at symbolic (j, k, q, η) (companion II, checks R6a–d, R7, R8). ■

REMARK 1 (WHY THE MIDDLE INDEX NEEDS TWO RULES). The physical middle index ℓ' is the pair $\ell'_1 = \ell'_2$. Raising it means applying (G1- ℓ'_1) then (G1- ℓ'_2) (in either order); intermediate elements are off-diagonal — that is the origin, and the resolution, of the off-diagonal companions in (M). At the level of this calculus the off-diagonal family is not a complication but the natural state space.

4 G2 — symmetries

THEOREM G2. (a) Reflection. ${}^{(j,k)}\mathcal{X}_{\ell;\ell'_1,\ell'_2;\ell''}^m(q) = {}^{(-k,-j)}\mathcal{X}_{\ell'';\ell'_2,\ell'_1;\ell}^m(-3-q)$ — on the physical slice, $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q) = \hat{\mathcal{D}}_{\ell''\ell'\ell}(-3-q)$ (N1). **(b) Parity.** $\mathcal{X}(-\beta) = (-1)^{\ell+\ell'_1+\ell'_2+\ell''} \mathcal{X}(\beta)$; on J -forms, $\eta \rightarrow -\eta$ flips $J_w \rightarrow -J_w$ and $p \rightarrow -p$. **(c) Gauge.** ${}^{(j+1,k+1)}\mathcal{X}(q) = {}^{(j,k)}\mathcal{X}(q+1)$: the lattice diagonal is a shift of the power-law exponent; only $j-k$ is a genuinely new label.

Proof. (a) Transpose both kernel factors (companion I, Lemma 3) and relabel: the $+\beta$ factor (ℓ, ℓ'_1) at weight $-1-j-q$ becomes a $-\beta$ factor (ℓ'_1, ℓ) at weight $3+j+q = -k^* - q^*$, and vice versa, with $q^* = -3-q$, $(j^*, k^*) = (-k, -j)$. (b) Kernel parity per factor (companion II, R1), phases multiply. (c) Both weights shift by one = both kernels evaluated at exponent $q+1$. (Exact checks G-V6, G-V8, G-V7 on generated elements.) ■

5 G3 — the Generation Theorem

THEOREM G3 (GENERATION FROM THE SEED). For $m = 0$: every element ${}^{(j,k)}\mathcal{X}_{\ell,\ell'_1,\ell'_2,\ell''}(q)$ — in particular every physical $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q)$ — is a finite linear combination, with coefficients in $\mathbb{Q}\left[\gamma, \frac{1}{p}\right][\{C_\ell\}]$, of seed elements (Go), computed by iterating the four rules (G1). Explicitly:

$$\hat{\mathcal{D}}_{\ell\ell'\ell''}(q) \in \frac{1}{4\gamma p^{2+\ell+2\ell'+\ell''}} \mathbb{Q}[\gamma, p]\text{-span}\left\{ J_{a+q} J_{b+q} : a, b \in \mathbb{Z} \right\},$$

and the recursion terminates in at most $O((\ell+\ell'+1)(\ell''+\ell'+1))$ rule applications, touching lattice points only inside the rectangle $[-\ell'_1, \ell] \times [-\ell'', \ell'_2]$ around (j, k) .

Proof. Termination and well-foundedness: each rule strictly lowers the raised index by 1 and 2 and leaves the others fixed; the companion terms move one lattice step; induction on $\ell + \ell'_1 + \ell'_2 + \ell''$. *Base:* all indices 0 is Theorem Go at the required lattice points. *Correctness:* each rule is an exact identity (Theorem G1). The p -power counting follows because each application of a rule multiplies by $1/p$ at worst once per raised unit, and the seed carries $1/(4\gamma p^2)$; γ -powers reduce via $\gamma^2 = 1 + p^2$. The window bound holds because raising ℓ steps east only, ℓ'_1 west only, ℓ'_2 north only, ℓ'' south only. (Implementation: `doppler_generator.wl`, functions `xg/dg`; exactness against the defining operator: checks **G-V1** — all $4^4 = 256$ elements with indices ≤ 3 at $(0, 0)$, symbolic q and symbolic rapidity — and **G-V2** at shifted lattice points.) ■

COROLLARY G4 (FACTORIZATION). Every element generated this way factorizes,

$${}^{(j,k)}\mathcal{X}_{\ell,\ell'_1,\ell'_2,\ell''}(q) = \frac{1}{\gamma} F_{\ell\ell'_1}^{(j)} \cdot G_{\ell'_2\ell''}^{(k)},$$

where F and G are each a $\mathbb{Q}[\gamma, p, \frac{1}{p}]$ -combination of single J 's: rules (G1- ℓ), (G1- ℓ'_1) touch only the first factor and (G1- ℓ'_2), (G1- ℓ'') only the second. Hence every diagonal $\hat{\mathcal{D}}_{\ell\ell'\ell''}$ closed form is a *product of two univariate J -brackets* over $4\gamma p^{2+\sigma}$ — the structure observed case-by-case on the site $(\hat{\mathcal{D}}_{101}, \hat{\mathcal{D}}_{020}, \hat{\mathcal{D}}_{202}, \dots)$ is now a theorem.

ALGORITHM (ANY $\hat{\mathcal{D}}_{HK'L}$ FROM $\hat{\mathcal{D}}_{000}$).

```
X(j,k; 0,0,0,0) = J[2+j+q] J[1+k+q] / (4 γ p²) # G0 seed = D000 shifted
X(j,k; ℓ,ℓ₁',ℓ₂',ℓ'') : # apply first applicable rule
  ℓ > 0 : +1/(C_ℓ p) (γ X[ℓ-1] - X^{(j+1,k)}[ℓ-1]) - (C_{ℓ-1}/C_ℓ) X[ℓ-2]
  ℓ₁' > 0 : -1/(C_ℓ₁' p) (γ X[ℓ₁'-1] - X^{(j-1,k)}[ℓ₁'-1]) - (C_{ℓ₁'-1}/C_ℓ₁') X[ℓ₁'-2]
  ℓ₂' > 0 : -1/(C_ℓ₂' p) (γ X[ℓ₂'-1] - X^{(j,k+1)}[ℓ₂'-1]) - (C_{ℓ₂'-1}/C_ℓ₂') X[ℓ₂'-2]
  ℓ'' > 0 : +1/(C_ℓ'' p) (γ X[ℓ''-1] - X^{(j,k-1)}[ℓ''-1]) - (C_{ℓ''-1}/C_ℓ'') X[ℓ''-2]
D(h,k',1) = X(0,0; h,k',k',1) # memoize; order of rules irrelevant
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6 Po–P4 — relations among *physical* Doppler operators only

Theorems G1–G3 still raise indices through the auxiliary family $\mathcal{X}$ (lattice companions, off-diagonal middles) — as do the site's $N_5/N_6/(M)$. This section eliminates the auxiliary family entirely. The key is the gauge dictionary of G2c: a lattice move composed with a diagonal shift is a shift of the power-law exponent, so the east companion of the ℓ -ladder is the south companion of the ℓ'' -ladder at $q + 1$: ${}^{(1,0)}\mathcal{X}(q) = {}^{(0,-1)}\mathcal{X}(q + 1)$. Subtracting the two ladders cancels it. In the statements below **every term is a physical element** $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q + n)$, $n \in \mathbb{Z}$; in operator language $q \rightarrow q + n$ is the conjugation $\nu^{-n}\hat{\mathcal{D}}\nu^n$, i.e. $\hat{\mathcal{O}}_\nu \rightarrow \hat{\mathcal{O}}_\nu - n$.

THEOREM P1 (OUTER PAIR RELATION; ANY M).

$$C_{\ell+1}^m \hat{\mathcal{D}}_{\ell+1,\ell'\ell''}(q) + C_{\ell}^m \hat{\mathcal{D}}_{\ell-1,\ell'\ell''}(q) - C_{\ell+1}^m \hat{\mathcal{D}}_{\ell,\ell',\ell''+1}(q+1) - C_{\ell}^m \hat{\mathcal{D}}_{\ell,\ell',\ell''-1}(q+1) = \frac{\gamma}{p} [\hat{\mathcal{D}}_{\ell\ell'\ell''}(q) - \hat{\mathcal{D}}_{\ell\ell'\ell''}(q+1)]$$

Operator form: $C_{\ell+1} \hat{\mathcal{D}}_{\ell+1} + C_{\ell} \hat{\mathcal{D}}_{\ell-1} - \nu^{-1} [C_{\ell+1} \hat{\mathcal{D}}_{\ell+1} + C_{\ell} \hat{\mathcal{D}}_{\ell-1}] \nu = \frac{\gamma}{p} (\hat{\mathcal{D}} - \nu^{-1} \hat{\mathcal{D}} \nu)$. It is self-reflected under N1. (Check **P1**: $m \leq 1$, indices ≤ 3 , symbolic (q, η) .)

THEOREM P0 (AXIS LADDERS; $M = 0$). On the two outer axes the companion resolves with an explicit scalar coefficient:

$$C_{a+1} \hat{\mathcal{D}}_{a+1,0,0}(q) + C_a \hat{\mathcal{D}}_{a-1,0,0}(q) = \frac{\gamma}{p} \hat{\mathcal{D}}_{a00}(q) - \frac{1}{p} \frac{J_{1+q}}{J_{2+q}} \hat{\mathcal{D}}_{a00}(q+1),$$

$$C_{b+1} \hat{\mathcal{D}}_{0,0,b+1}(q) + C_b \hat{\mathcal{D}}_{0,0,b-1}(q) = \frac{\gamma}{p} \hat{\mathcal{D}}_{00b}(q) - \frac{1}{p} \frac{J_{2+q}}{J_{1+q}} \hat{\mathcal{D}}_{00b}(q-1).$$

Solved for the top element, each is an *explicit one-index raising relation seeded by $\hat{\mathcal{D}}_{000}$ alone*: e.g. $C_1 \hat{\mathcal{D}}_{100}(q) = \frac{\gamma}{p} \hat{\mathcal{D}}_{000}(q) - \frac{1}{p} \frac{J_{1+q}}{J_{2+q}} \hat{\mathcal{D}}_{000}(q+1)$. (Checks **P0/P0'**.)

THEOREM P2 (RANK-ONE; ANY M). At fixed middle index and fixed q , the doubly-indexed array of physical elements is rank one: for all h_1, h_2, l_1, l_2 ,

$$\hat{\mathcal{D}}_{h_1\ell'l_1}(q) \hat{\mathcal{D}}_{h_2\ell'l_2}(q) = \hat{\mathcal{D}}_{h_1\ell'l_2}(q) \hat{\mathcal{D}}_{h_2\ell'l_1}(q), \quad \text{so} \quad \hat{\mathcal{D}}_{h\ell'l} = \frac{\hat{\mathcal{D}}_{h\ell'0} \hat{\mathcal{D}}_{0\ell'l}}{\hat{\mathcal{D}}_{0\ell'0}}.$$

One row and one column at each ℓ' determine everything. (Check **P2**: all 2×2 minors, $m \leq 1$.)

THEOREM P3 (MIDDLE INDEX). (a) The bracket family of G4b is itself a ratio of physical elements,

$$F_{ab}(q) = (-1)^b \frac{J_q \hat{\mathcal{D}}_{ab0}(q)}{2p \hat{\mathcal{D}}_{b00}(q-1)} \quad (\text{check } \texttt{\textbf{P3a}}),$$

so every statement of the F -calculus is a statement about physical $\hat{\mathcal{D}}$'s. (b) Substituting (a) into the product of the two bracket ladders gives the pure-physical middle-index step: $\hat{\mathcal{D}}_{h,\ell'+1,0}(q)$ equals a nine-term combination of $\hat{\mathcal{D}}_{h,\ell',0}, \hat{\mathcal{D}}_{h,\ell'-1,0}$ (at $q, q-1$) with coefficients built from γ, p, J -ratios and the axis elements $\hat{\mathcal{D}}_{\ell'00}, \hat{\mathcal{D}}_{\ell'-1,0,0}$ (verified exactly, check **P3b**). The step is necessarily *nonlinear* (ratios of physical elements): a linear pure-physical middle recursion cannot exist, because a shift of the shared index moves both factors and the diagonal slice is codimension one in the family it generates — the rank-one structure (P2) is precisely what survives of the product structure at the physical level.

COROLLARY P4 (PURE-PHYSICAL GENERATION FROM $\hat{\mathcal{D}}_{000}$). Every $\hat{\mathcal{D}}_{\ell\ell'\ell''}(q)$ is computed from $\hat{\mathcal{D}}_{000}$ with every intermediate quantity a physical Doppler element: P0 generates the axis families $\hat{\mathcal{D}}_{a00}(q), \hat{\mathcal{D}}_{00b}(q)$; P3b climbs the middle index along the border $\hat{\mathcal{D}}_{h\ell'0}$; reflection (G2a) gives $\hat{\mathcal{D}}_{0\ell'l}$; and rank-one (P2) fills in all remaining elements. The $\hat{\mathcal{O}}$ -shifts $q \rightarrow q \pm 1$ that appear throughout are not an artifact: by the gauge dictionary they are the physical remnant of the weight lattice, and P1 shows the residual content of the two outer ladders after the lattice is eliminated.

7 G5 — the rapidity flow at the Doppler level (for completeness)

The generation scheme above is purely algebraic. The differential structure, restated on the same family with the corrected lattice coefficients (companion I, §8.3):

THEOREM G5A (OUTER FLOW = N3, LATTICE FORM).

$$\begin{aligned} \partial_{\eta}^{(j,k)} \mathcal{X} = & -(\ell + 3 + j - \hat{O}_{\nu}) C_{\ell+1}^m \mathcal{X}[\ell+1] + (\ell - 2 - j + \hat{O}_{\nu}) C_{\ell}^m \mathcal{X}[\ell-1] \\ & - (\ell'' - k + \hat{O}_{\nu}) C_{\ell''+1}^m \mathcal{X}[\ell''+1] + (\ell'' + 1 + k - \hat{O}_{\nu}) C_{\ell''}^m \mathcal{X}[\ell''-1] - \beta \mathcal{X}, \end{aligned}$$

middle indices and lattice point fixed: the generator of the whole family in rapidity.

THEOREM G5B (MIRROR FLOW — NEW). The same derivative can instead be written moving *only the middle indices*:

$$\begin{aligned} \partial_{\eta}^{(j,k)} \mathcal{X} = & +(\ell'_1 - 1 - j + \hat{O}_{\nu}) C_{\ell'_1+1}^m \mathcal{X}[\ell'_1+1] - (\ell'_1 + 2 + j - \hat{O}_{\nu}) C_{\ell'_1}^m \mathcal{X}[\ell'_1-1] \\ & + (\ell'_2 + 2 + k - \hat{O}_{\nu}) C_{\ell'_2+1}^m \mathcal{X}[\ell'_2+1] - (\ell'_2 - 1 - k + \hat{O}_{\nu}) C_{\ell'_2}^m \mathcal{X}[\ell'_2-1] - \beta \mathcal{X}, \end{aligned}$$

with outer indices and lattice point spectating (exact check **G-V9**, symbolic q). The difference of G5a and G5b is the algebraic cross-slot balance (companion II, R11) applied to each factor. On the diagonal slice G5b does not close — the mirror statement of ℓ' being a spectator of N3.

8 G6 — general m

THEOREM G6. For any azimuthal sector $m \geq 0$, the rules (G1), symmetries (G2) and both flows (G5) hold verbatim with C_{ℓ}^m (checks in companion II were run at $m \leq 1$, kernel-level masters at $m \leq 2$, all symbolic weights). Only the seed changes: the base case sits at all indices $= m$,

$$\partial_{\eta}^{(j,k)} \mathcal{X}_{m;m,m;m}^m(q) = \frac{1}{\gamma} \kappa_{-1-j-q}^m \kappa_{-k-q}^m, \quad \kappa_d^m \equiv {}^d\mathcal{K}_{mm}^m = \gamma^{-(m+d)} {}_2F_1\left(\frac{m+d}{2}, \frac{m+d+1}{2}; \frac{3}{2} + m; \beta^2\right),$$

which for $m = 0$ reduces to (Go) via the classical ${}_2F_1$ evaluation $\kappa_d^0 = J_{1-d}/(2p)$ -type identities. Everything above m is again pure ladder algebra.

9 Catalog of closed forms

Produced by the generator and reduced mod $\gamma^2 = 1 + p^2$ (shorthand $J_n \equiv J_{n+q}$; $\hat{O}_{\nu} \rightarrow -q$; all verified exactly against the defining operator):

COROLLARY G4B (ONE BRACKET FAMILY SUFFICES). Define the bracket family purely inside the calculus,

$$F_{hk'}(q) \equiv \frac{2p\gamma \mathcal{X}_{h;k',0;0}(q)}{J_{1+q}}, \quad F_{00} = \frac{J_{2+q}}{2p}$$

(the denominator is the trailing seed factor; equivalently $F_{hk'} = {}^{-1-q}\mathcal{K}_{hk'}(+\beta)$). Then, combining G4 with parity and gauge,

$$\boxed{\hat{\mathcal{D}}_{hk'l}(q) = \frac{(-1)^{k'+l}}{\gamma} F_{hk'}(q) F_{k'l}(q-1)}$$

— every Doppler element is a bilinear form in the single one-parameter family F , which itself obeys the two three-term ladders (G1- ℓ), (G1- ℓ'_1) with $F \rightarrow F(q \mp 1)$ companions, and the bracket-level reflection $F_{hk'}(q) = (-1)^{h+k'} F_{k'h}(-4-q)$.

Bracket table (shorthand $J_n \equiv J_{n+q}$; verified exactly, full table to indices ≤ 3 in `out_doppler_brackets.txt`; arbitrary indices by the algorithm):

$$\begin{aligned}
F_{00} &= \frac{J_2}{2p}, & F_{01} &= \frac{\sqrt{3}(J_1 - \gamma J_2)}{2p^2}, & F_{10} &= \frac{\sqrt{3}(\gamma J_2 - J_3)}{2p^2}, \\
F_{02} &= \frac{\sqrt{5}(3J_0 - 6\gamma J_1 + (3+2p^2)J_2)}{4p^3}, & F_{20} &= \frac{\sqrt{5}((3+2p^2)J_2 - 6\gamma J_3 + 3J_4)}{4p^3}, \\
F_{11} &= \frac{3(\gamma J_1 - (2+p^2)J_2 + \gamma J_3)}{2p^3}, \\
F_{12} &= \frac{\sqrt{15}(3\gamma J_0 - 3(3+2p^2)J_1 + \gamma(9+2p^2)J_2 - (3+2p^2)J_3)}{4p^4}, \\
F_{21} &= \frac{\sqrt{15}((3+2p^2)J_1 - \gamma(9+2p^2)J_2 + 3(3+2p^2)J_3 - 3\gamma J_4)}{4p^4}, \\
F_{22} &= \frac{5(3(3+2p^2)J_0 - 12\gamma(3+p^2)J_1 + 2(27+24p^2+2p^4)J_2 - 12\gamma(3+p^2)J_3 + 3(3+2p^2)J_4)}{8p^5}.
\end{aligned}$$

Assembled examples. $\hat{\mathcal{D}}_{000} = \frac{1}{\gamma}F_{00}(q)F_{00}(q-1) = \frac{J_2J_1}{4\gamma p^2}$ (the seed);
 $\hat{\mathcal{D}}_{101} = -\frac{1}{\gamma}F_{10}(q)F_{01}(q-1) = \frac{3}{4\gamma p^4}(\gamma J_2 - J_3)(\gamma J_1 - J_0)$ and $\hat{\mathcal{D}}_{020} = \frac{1}{\gamma}F_{02}(q)F_{20}(q-1)$, both
matching the site exactly (G-V3/4); the "missing element" of the kinematic-dipole sector (site N16),

$$\hat{\mathcal{D}}_{121}(q) = -\frac{15}{16\gamma p^8} \left[3\gamma J_0 - 3(3+2p^2)J_1 + \gamma(9+2p^2)J_2 - (3+2p^2)J_3 \right] \left[(3+2p^2)J_0 - \gamma(9+2p^2)J_1 + 3(3+2p^2)J_2 \right]$$

manifestly N1a-invariant (the brackets swap up to sign under $q \rightarrow -3 - q$, using $J_{-w} = J_w$). The generator also certifies elements with no published closed form, e.g. $\hat{\mathcal{D}}_{212}$, $\hat{\mathcal{D}}_{222}$, $\hat{\mathcal{D}}_{303}$, $\hat{\mathcal{D}}_{112}$ (see `out_doppler_generator.txt`).

The $\hat{\mathcal{D}}_{121}$ form combines with $\hat{\mathcal{D}}_{101}$ into the kinematic-dipole operator $M_{11} = \langle \hat{\mathcal{D}}_{101} \rangle + \frac{1}{10} \langle \hat{\mathcal{D}}_{121} \rangle$ (site N16); the elements $\hat{\mathcal{D}}_{212}$, $\hat{\mathcal{D}}_{222}$, $\hat{\mathcal{D}}_{303}$, $\hat{\mathcal{D}}_{112}$ do not appear on the site — they are new, produced and certified by the generator.

10 Verification record

Check	Statement	Symbolic in	Range	Result
G-V1	generator = defining operator	q, η	all 4^4 index tuples ≤ 3 , $(j, k) = (0, 0)$	PASS (exact)
G-V2	same at lattice points $(\pm 1, \mp 1)$	q, η	indices ≤ 2	PASS (exact)
G-V3–5	$\hat{\mathcal{D}}_{101}, \hat{\mathcal{D}}_{020}, \hat{\mathcal{D}}_{202}$ vs site closed forms	q, η	—	PASS (exact)
G-V6	reflection (G2a) on generated elements	q, η	$h, k', l \leq 2$	PASS (exact)
G-V7	gauge (G2c)	q, η	indices ≤ 2	PASS (exact)
G-V8	parity (G2b)	q, η	indices ≤ 2	PASS (exact)
G-V9	mirror rapidity ODE (G5b, new)	q, η	$(j, k) \in \{0, 1\}^2$, indices ≤ 2	PASS (exact)
P1	outer pair relation (pure physical)	q, η	$m \leq 1$, indices ≤ 3	PASS (exact)
Po/Po'	axis ladders, J-ratio coefficients	q, η	$a, b \leq 3$	PASS (exact)

P2	rank-one minors	q, η	$m \leq 1$, indices ≤ 3	PASS (exact)
P3a/P3b	bracket-as-ratio; pure-D middle step	q, η	$a, b \leq 3$; $h \leq 2, \ell' \leq 2$	PASS (exact)

Plus `doppler_brackets.wl`: the $G4/G4b$ factorization $\hat{D}_{hk'l} = \frac{(-1)^{k'+l}}{\gamma} F_{hk'}(q) F_{k'l}(q-1)$ verified exactly for all indices ≤ 3 ; and `pure_physical_relations.wl`: Theorems P0–P3 (`log out_pure_physical_relations.txt`). Runtimes: generator harness 169 s; brackets 40 s; pure-physical 41 s (Wolfram 15). Every check is a rational-function identity in $(e^\eta, e^{q\eta})$ reduced to literal 0.

References

1. Companion I: *The Rapidity ODE for the Doppler Operator: An Explicit Proof* — conventions, Lemmas 1–4, N3, lattice-ODE correction.
2. Companion II: *Index Recurrences for the Doppler Operator* — Master Lemma, Theorem U, A.7b correction, analytic N4.
3. [RC26], [CR26], [CR25], [HLC26] — as in the companions.
4. The Doppler Operator Project — [site](#): N7 seeds, N8 closed forms, N10 middle index, N16 kinematic dipole.

Prepared with assistance from Claude (Anthropic). All symbolic checks machine-verified with Wolfram Language; the generator and every identity in this document are executable via the companion script.